

Zero Quotient Dimension Can Require Continuum Many Subideal Generators

A ZFC counterexample to Question 5 of *Subideals of Operators II*

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Status: candidate counterexample, likely valid, pending human review. The construction gives a proper $K(H)$ -ideal with zero quotient dimension but generator number exactly the continuum. It refutes both sentences of the source's Question 5 under their literal hypotheses.

1 Result at a glance

Let H be a separable infinite-dimensional complex Hilbert space. Patnaik and Weiss define, for a J -ideal $\mathcal{I}$,

$$\mathcal{I}^0 = \text{span}(\mathcal{I}J + J\mathcal{I}) + J(\mathcal{I})J,$$

where $(\mathcal{I})$ is the $B(H)$ -ideal generated by $\mathcal{I}$ [1]. They ask whether $\dim_{\mathbb{C}}(\mathcal{I}/\mathcal{I}^0)$ is the cardinality of a generating set, or at least whether dimension below $\mathfrak{c} = 2^{\aleph_0}$ forces a generating set of cardinality below $\mathfrak{c}$.

We construct a proper $B(H)$ -ideal

$$\mathcal{I}_* \subsetneq K(H)$$

such that

$$\mathcal{I}_*^2 = \mathcal{I}_*, \quad \min\{|S| : (S)_{K(H)} = \mathcal{I}_*\} = \mathfrak{c}.$$

Taking the ambient ideal $J = K(H)$ and $\mathcal{I} = \mathcal{I}_*$, idempotence gives

$$\mathcal{I} = \mathcal{I}^2 \subset \mathcal{I}J \subset \mathcal{I},$$

so $\mathcal{I}^0 = \mathcal{I}$. Therefore

$$\dim_{\mathbb{C}}(\mathcal{I}/\mathcal{I}^0) = 0 < \mathfrak{c} \quad \text{while} \quad \min\{|S| : (S)_J = \mathcal{I}\} = \mathfrak{c}.$$

2 Source question

The questions appear in Section 6, page 9 of arXiv:1209.6323 [1]. The following is a full-width crop of the source page.

QUESTIONS

1. Determine which J -ideals have generating sets of cardinality less than $\mathfrak{c}$?
2. Do J -ideals have minimal cardinality generating sets?
3. Traces. $B(H)$ -ideals are important in part because of the importance of their traces, unitarily invariant functionals on $B(H)$ -ideals. Because $B(H)$ -ideals J contain no unitaries this concept does not apply. Is there a modification for which one obtains a useful concept of traces for J -ideals?
4. The study of commutators in $B(H)$ -ideals is directly related to their traces. What can be said about the commutator structure of the J -ideals that can be used to motivate notions of traces for J -ideals?
5. Is the dimension of $\mathcal{I}/\mathcal{I}^0$ the cardinality of some generating set for $\mathcal{I}$?
At least if $\dim \mathcal{I}/\mathcal{I}^0 < \mathfrak{c}$, must $\mathcal{I}$ possess a generating set of cardinality less than $\mathfrak{c}$?
(See also Question [4.3](#))

Figure 1. The source's five questions. This packet answers Question 5.

Question 5 (transcription). “Is the dimension of $\mathcal{I}/\mathcal{I}^0$ the cardinality of some generating set for $\mathcal{I}$? At least if $\dim \mathcal{I}/\mathcal{I}^0 < \mathfrak{c}$, must $\mathcal{I}$ possess a generating set of cardinality less than $\mathfrak{c}$?”

3 Definitions and a finite-domination lemma

For a subset S of a $B(H)$ -ideal J , write (S) for the $B(H)$ -ideal generated by S , and $(S)_J$ for the two-sided ideal of the nonunital ring J generated by S . Let c_0^* be the cone of positive nonincreasing null sequences. For $\xi \in c_0^*$, put

$$D_\xi = \text{diag}(\xi_1, \xi_2, \dots).$$

For $m \geq 1$, the m -fold ampliation of a sequence is

$$(D_m \xi)_n = \xi_{\lceil n/m \rceil}.$$

The proof uses the following elementary form of the characteristic-set criterion. We include the argument to make the packet independent of a classification theorem.

Lemma 1 (Finite ideal domination). *Let $\xi, \eta_1, \dots, \eta_q \in c_0^*$. If*

$$D_\xi \in (D_{\eta_1}, \dots, D_{\eta_q}),$$

then there are $C > 0$ and positive integers m_i such that

$$\xi \leq C \sum_{i=1}^q D_{m_i} \eta_i \quad \text{coordinatewise.}$$

Proof. Membership in the algebraically generated ideal expresses D_ξ as a finite sum

$$D_\xi = \sum_{j=1}^r A_j D_{\eta_{i_j}} B_j, \quad A_j, B_j \in B(H).$$

For $T_j = A_j D_{\eta_{i_j}} B_j$,

$$s_n(T_j) \leq \|A_j\| \|B_j\| (\eta_{i_j})_n.$$

Put $a = \lceil n/r \rceil$. Then $ra - r + 1 \leq n$, and the repeated Ky Fan singular-number inequality [2, Chapter 2] gives

$$\begin{aligned} \xi_n = s_n(D_\xi) &\leq s_{ra-r+1} \left(\sum_{j=1}^r T_j \right) \\ &\leq \sum_{j=1}^r s_a(T_j) \leq \sum_{j=1}^r \|A_j\| \|B_j\| (\eta_{i_j})_{\lceil n/r \rceil}. \end{aligned}$$

Grouping equal η_i 's and enlarging one constant yields the stated finite ampliation bound. $\square$

4 Proof intuition

The quotient $\mathcal{I}/\mathcal{I}^0$ sees generators only modulo products with the ambient ideal. An idempotent ideal can therefore be completely invisible to this quotient: if $\mathcal{I}^2 = \mathcal{I}$, then $\mathcal{I}K(H) = \mathcal{I}$ and the quotient is zero.

The difficulty is to make such an ideal require *continuum many* generators in ZFC. We encode a continuum independent family of subsets of $\mathbb{N}$ into compact diagonal operators. At infinitely many enormous blocks, one selected sequence is relatively large while any prescribed finite collection of the other sequences is extremely small. Lemma 1 then prevents the selected diagonal from lying in the ideal generated by those competitors.

Finally, we include every successive square root of every diagonal generator. Each generator is then literally the square of another generator, so the resulting ideal is algebraically idempotent. The independent-family separation survives all these roots and forces the generator number to be $\mathfrak{c}$.

5 The counterexample

5.1 A continuum independent family

Lemma 2. *There is a family*

$$\mathcal{A} = \{A_\alpha : \alpha < \mathfrak{c}\} \subset \mathcal{P}(\mathbb{N})$$

such that every finite Boolean cell is infinite. Explicitly, for distinct $\alpha_0, \dots, \alpha_q$ and arbitrary $\varepsilon_i \in \{0, 1\}$, the set

$$\{k : \mathbf{1}_{A_{\alpha_i}}(k) = \varepsilon_i \text{ for } 0 \leq i \leq q\}$$

is infinite.

Proof. Let

$$\Omega = \bigcup_{n \geq 1} (\{n\} \times \mathcal{P}(2^n)),$$

where 2^n denotes the finite set of binary strings of length n . The set Ω is countable. For $x \in 2^{\mathbb{N}}$, define

$$A_x = \{(n, F) \in \Omega : x|_n \in F\}.$$

Given finitely many distinct x_i , their length- n prefixes are distinct for every sufficiently large n . For a prescribed Boolean pattern, let $F \subset 2^n$ contain exactly those prefixes assigned the value 1. The point (n, F) realizes the pattern for every sufficiently large n . Identify Ω with $\mathbb{N}$. $\square$

5.2 Strongly incomparable null sequences

Set

$$N_0 = L_0 = 1, \quad N_k = L_k = 2^{2^k} \quad (k \geq 1).$$

For $A \in \mathcal{A}$, define

$$r_A(k) = \max(\{0\} \cup (A \cap \{1, \dots, k\})).$$

On the block $N_k \leq n < N_{k+1}$, put

$$f_A(n) = L_{r_A(k)}, \quad \xi_A(n) = e^{-f_A(n)}.$$

The values before the first nontrivial block are set equal to e^{-L_0} . Since A is infinite, f_A is nondecreasing and tends to infinity. Thus $\xi_A \in c_0^*$.

Lemma 3 (Strong separation). *For each $A \in \mathcal{A}$,*

$$D_{\xi_A} \notin (D_{\xi_B^t} : B \in \mathcal{A} \setminus \{A\}, t > 0).$$

Proof. Ideal membership uses only finitely many displayed generators. If the conclusion failed, Lemma 1 would give distinct $B_1, \dots, B_q \neq A$, positive t_i , integers m_i , and $C > 0$ such that

$$\xi_A \leq C \sum_{i=1}^q D_{m_i}(\xi_{B_i}^{t_i}). \quad (1)$$

Let $M = \max_i m_i$ and $t = \min_i t_i > 0$. Lemma 2 gives infinitely many k for which

$$k \notin A, \quad k \in B_i \quad (1 \leq i \leq q).$$

Choose such a sufficiently large k , and set $n = MN_k$. Since $N_{k+1}/N_k \rightarrow \infty$, the integers n and $\lceil n/m_i \rceil$ all lie in the k -th block. Consequently

$$f_A(n) \leq L_{k-1}, \quad f_{B_i}(\lceil n/m_i \rceil) = L_k.$$

At coordinate n , the ratio of the right side of (1) to its left side is at most

$$Cq \exp(-tL_k + L_{k-1}) \longrightarrow 0.$$

This contradicts (1). $\square$

5.3 Idempotence, properness, and generator number

Define

$$\mathcal{G} = \{D_{\xi_A^{2^{-r}}} : A \in \mathcal{A}, r \in \mathbb{N}_0\}, \quad \mathcal{I}_* = (\mathcal{G}).$$

Every generator is compact, so $\mathcal{I}_* \subset K(H)$.

Lemma 4. *The ideal $\mathcal{I}_*$ satisfies:*

- (i) $\mathcal{I}_* \subsetneq K(H)$;
- (ii) $\mathcal{I}_*^2 = \mathcal{I}_*$;
- (iii) every $B(H)$ -generating set for $\mathcal{I}_*$ has cardinality at least $\mathfrak{c}$.

Proof. For (i), let

$$\zeta_n = \frac{1}{\log(n+1)}.$$

This belongs to c_0^* . If $D_\zeta \in \mathcal{I}_*$, finite ideal membership and Lemma 1 would bound ζ by finitely many ampliations of $\xi_{B_i}^{t_i}$. Choose infinitely many $k \in \bigcap_i B_i$, which Lemma 2 supplies, and again evaluate at $n = MN_k$. With $t = \min_i t_i$, the ratio of the proposed right side to ζ_n is at most

$$Cq \log(MN_k + 1)e^{-tL_k} \longrightarrow 0.$$

Thus $D_\zeta \notin \mathcal{I}_*$, proving strict containment.

For (ii), every generator has the exact factorization

$$D_{\xi_A^{2^{-r}}} = \left(D_{\xi_A^{2^{-(r+1)}}} \right)^2.$$

Therefore $\mathcal{G} \subset \mathcal{I}_*^2$. Since $\mathcal{I}_*^2$ is a $B(H)$ -ideal containing $\mathcal{G}$, minimality gives $\mathcal{I}_* \subset \mathcal{I}_*^2$. The reverse inclusion is automatic for every ideal.

For (iii), suppose that $S \subset \mathcal{I}_*$, $|S| < \mathfrak{c}$, and $(S) = \mathcal{I}_*$. Every element $T \in S$ belongs to the ideal generated by a finite subset of $\mathcal{G}$. Let $U \subset \mathcal{A}$ be the union of the corresponding finite sets of A -indices. Then $|U| < \mathfrak{c}$; no regularity assumption on $\mathfrak{c}$ is involved, since a union of κ finite sets has cardinality at most κ for every infinite cardinal κ .

Choose $A \in \mathcal{A} \setminus U$. The ideal (S) is contained in the ideal generated by

$$\{D_{\xi_B^{2^{-r}}} : B \in U, r \in \mathbb{N}_0\}.$$

Lemma 3 says that this ideal does not contain D_{ξ_A} , contradicting $(S) = \mathcal{I}_*$. □

Theorem 5 (Negative answer to Question 5). *There is a proper $K(H)$ -ideal $\mathcal{I} \subsetneq K(H)$ such that*

$$\dim_{\mathbb{C}}(\mathcal{I}/\mathcal{I}^0) = 0 < \mathfrak{c}, \quad \min\{|S| : (S)_{K(H)} = \mathcal{I}\} = \mathfrak{c}.$$

Hence both assertions asked in Section 6, Question 5 of arXiv:1209.6323 have negative answers in ZFC.

Proof. Take $\mathcal{I} = \mathcal{I}_*$ and the ambient ideal $J = K(H)$. If $(S)_{K(H)} = \mathcal{I}_*$, taking the $B(H)$ -ideal hull of both sides gives

$$(S) = \mathcal{I}_*.$$

Lemma 4(iii) therefore yields $|S| \geq \mathfrak{c}$.

For the matching upper bound, idempotence gives $\mathcal{I}_*^3 = \mathcal{I}_*$. Since $\mathcal{I}_* \subset K(H)$,

$$\mathcal{I}_* = \mathcal{I}_*^3 \subset K(H) \mathcal{I}_* K(H) = K(H)(\mathcal{G})K(H) \subset (\mathcal{G})_{K(H)} \subset (\mathcal{G}) = \mathcal{I}_*.$$

Thus $(\mathcal{G})_{K(H)} = \mathcal{I}_*$. The family $\mathcal{G}$ has cardinality $\mathfrak{c}$, so the minimum is exactly $\mathfrak{c}$.

By Lemma 4(ii) and $\mathcal{I}_* \subset K(H)$,

$$\mathcal{I}_* = \mathcal{I}_*^2 \subset \mathcal{I}_* K(H) \subset \mathcal{I}_*.$$

Thus $\mathcal{I}_* K(H) = \mathcal{I}_*$. The defining expression for $\mathcal{I}^0$ contains $\mathcal{I}J = \mathcal{I}$, while it is always contained in $\mathcal{I}$. Hence $\mathcal{I}^0 = \mathcal{I}$, and the quotient has dimension zero. $\square$

6 Verification notes

The proof was subjected to the following checks.

1. The independent family is constructed explicitly on a countable set; no additional set-theoretic axiom is used.
2. Finite ideal membership is converted to ampliation domination using only a finite operator representation and the Ky Fan inequality.
3. The choice $n = MN_k$ handles all finitely many ampliations simultaneously and remains in the same block for sufficiently large k .
4. The separation lemma allows arbitrary positive powers, so adjoining all dyadic roots does not weaken the generator obstruction.
5. The slowly decreasing ζ proves $\mathcal{I}_* \subsetneq K(H)$; the counterexample is a proper subideal of its ambient ideal.
6. The equality $\mathcal{I}_*^2 = \mathcal{I}_*$ is algebraic: each generator is an exact product. No closure operation occurs.
7. A set of fewer than $\mathfrak{c}$ ideal elements involves fewer than $\mathfrak{c}$ original indices because each element uses only finitely many algebraic generators.
8. Taking $B(H)$ -ideal hulls of $(S)_{K(H)} = \mathcal{I}_*$ gives $(S) = \mathcal{I}_*$, so the ordinary generator lower bound transfers to subideal generation. Conversely, $\mathcal{I}_*^3 = \mathcal{I}_* \subset K(H)\mathcal{I}_*K(H)$ proves that the displayed continuum family really generates $\mathcal{I}_*$ as a $K(H)$ -ideal.

The reusable script `code/verify_block_estimates.py` evaluates the two logarithmic upper bounds for sample parameters. By block $k = 4$ they are already approximately $-3.8 \cdot 10^3$ and $-4.1 \cdot 10^3$. This is a scale-arithmetic consistency check only; it is not part of the proof.

7 Scope, limitations, and novelty check

Theorem 5 answers Question 5 only. It does not claim to settle Questions 1–4.

The example has zero quotient dimension. This is included in the literal wording of Question 5, and $\mathcal{I}$ is a proper subideal of the ambient $K(H)$. If the intended question had the additional unstated hypothesis $\dim(\mathcal{I}/\mathcal{I}^0) > 0$, that variant would remain open.

The bounded novelty search on 23 July 2026 covered:

- the run’s registry, solution, attempt, and proof-gap indexes;
- arXiv:1209.6323, its exact title, and exact phrases from Question 5;
- searches combining subideals, quotient dimension, generator cardinality, idempotent $B(H)$ -ideals, and continuum generators;
- the later Patnaik–Weiss survey arXiv:1303.5697 [3].

The survey develops subideal traces and commutator spaces, but no later source explicitly answering the generator-dimension Question 5 was located. This supports moderate novelty confidence, not an exhaustive claim of priority.

Human review recommendation. Send the packet to an operator-ideal expert. The decisive checks are Lemma 1, the finite-support argument in Lemma 4(iii), and the source convention for J -generation.

References

- [1] S. Patnaik and G. Weiss, *Subideals of operators II*, arXiv:1209.6323 (2012), <https://arxiv.org/abs/1209.6323>.
- [2] B. Simon, *Trace Ideals and Their Applications*, second edition, Mathematical Surveys and Monographs 120, American Mathematical Society, 2005.
- [3] S. Patnaik and G. Weiss, *A survey on subideals of operators and an introduction to subideal-traces*, arXiv:1303.5697 (2013), <https://arxiv.org/abs/1303.5697>.